# Lesson: Light sensor (Photoresistor) with Plot

**Big Picture**
This lesson will introduce the photoresistor while allowing students to become more acquainted with the BBC micro:bit microcontroller hardware and Makecode *Blocks Editor* software tool. The students will create a program that will display the brightness level on the micro:bit screen as a plot.

**Objectives**
Students will be able to:

- Define Variable
- Define Sensor
- Define photoresistor sensor

**Alabama Standards Alignment**

7 (Fifth Grade): Identify Variables.

- Examples: Determine if a variable is required for use later in the program.

8 (Fifth Grade): Demonstrate the programs require known starting values that may need to be updated appropriately during the execution of programs

- Examples: create a program that sets a variable to an initial value then later updates (changes) the value of the variable.

**Links to Resources**
Micro:bit Light Sensing: [https://learn.adafruit.com/micro-bit-lesson-4-sensing-light-and-temperature](https://learn.adafruit.com/micro-bit-lesson-4-sensing-light-and-temperature)

**Preparation**

- Light_sensor_with_brightness_plot_student_handout: Tutorial handout found on lesson page

Choose a presentation method:

- Instructor may lead the students through the lesson using the student handout
- Students may work at their own pace using the student handout. You may also post the video and tutorial locally and allow students to choose.

**Materials Required**
Each student (or pair of students) requires:

- Tutorial handout
- micro:bit
- USB cable
- 10K Ω resistor
- Photoresistor (LDR)

- Alligator to breadboard jumper wires (3)
- MakeCode editor
- Internet connected computer with modern browser
  **Note: Browsers known to work with micro:bit software includes Firefox, Chrome, Safari, and Microsoft Edge For a complete list, visit this page:* https://makecode.microbit.org/browsers
- Source of light, useful if you would like to test by changing brightness quickly (optional)

**Vocabulary and Concepts**

Variable: A named container for storing a value that may change while the program is running.

Sensor: An input device that reads or measures a physical property and converts it to a numerical value.

Light Sensor: A "photoresistor" sensor that measures the brightness of light as resistance

**Teaching Guide**

Getting started (10 mins)

Tell the class that they will create a micro:bit program with a Light Sensor today. Before they start programming, everyone needs to learn the new vocabulary terms.

Activity (40 mins)

The class is now ready to create their micro:bit with the sensor . Use your chosen method to demonstrate how to complete the activity. Make sure students are able to witness the plot change as the shade the photoresistor with their hands. Once students get a plot displayed on the screen, allow them time to experiment with the micro:bit with by covering and uncovering the sensor to change the plot. Encourage students to share their results with the class. It is important to build a sense of accomplishment early in CS Making so that students will be engaged quickly and are more likely to persevere when projects become more challenging.

Wrap Up (5 mins)

Review the 3-vocabulary words.

Variable: A named container for storing a value that may change while the program is running.

Sensor: An input device that reads or measures a physical property and converts it to a numerical value.

Light Sensor: a "photoresistor" sensor that measures the brightness of light as resistance